

UCE831 :BRIDGE ENGINEERING				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: The course introduces various components of bridges and their various types. Standard loading standards developed by IRC which form a consistent basis for design are introduced. The course provides a lucid exposition of the theory and design of, RC box culverts and T- Beam Bridges,. The course covers the theory and design of superstructure and substructure (piers and abutments), bearing and foundations.				
Introduction: Definition, components of a bridge, classifications, the importance of bridges, Need for investigations, selection of bridge site, preliminary data to be collected, sub-surface investigations for a bridge Standard Specifications: Road bridges, I.R.C. loadings, code provisions on width of carriageway, clearances Design of RC Culvert: Components, Design loadings, Design of slab culvert, Design of single cell box culvert RC T-beam Bridges: Introduction, <i>Design of Superstructure and Substructure</i> Design of Superstructure: Design of <i>Two-way Deck Slab</i> using Pigeaud's Curves, Courbon's Theory, Design of <i>Longitudinal and Cross Girders</i> Design of Substructure: Various types of expansion bearing and fixed bearings, Design of <i>Elastomeric Laminated bearings</i>, Design of <i>Abutment</i> Foundations: Overview, Design of Pile Cap, Design of Well Foundation				
Experimental Project/assignment/Micro Project Design of single cell box culvert for different classes of IRC loading using software				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: 1. Design various components of superstructure of T-beam bridges 2. Design various components of substructure and foundation of T-beam bridges 3. Design single cell box culvert on software				
Text Books 1. Victor, D. Johnson, Elements of Bridge Engineering, Oxford and IBH Publishers, New Delhi(2019) 2. Vazirani & Ratwani, Design of Concrete Bridges, Khanna Publishers, New Delhi (2010) 3. Raju, N. Krishna, Design of Bridges, 5th Edition, Oxford and IBH (2019)				
Reference Books 1. Raina, V.K., Analysis, Design and Construction of Bridges, Tata McGraw Hill (2010)				

2. Jagadeesh T.R., Jayaram M.A., Design of Bridge Structures, 3rd Edition, PHI (2020)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25